

CS504 Final Term Solved MCQs

(VISIT VURANK FOR MORE)

Question No: 1 (Marks: 1) Please choose one The two main documents produced during this phase are Requirement Statement and Requirement Specification. They are also called Requirement Definition and

▶ **Functional Specification**

▶ Mathematical specification

▶ System Specification

▶ None of the given

Question No: 2 (Marks: 1) - Please choose one

A decision table should be used

▶ To document all conditional statements

▶ To guide the development of the project management plan

▶ Only when building an expert system

▶ **when a complex set of conditions and actions appears in a component**

Question No: 3 (Marks: 1) Please choose one Which of the following is a tool used in requirement analysis?

▶ Flow Graphs

▶ **Data Flow Diagrams**

▶ Activity Networks

▶ Module Dependency Diagrams

Question No: 4 (Marks: 1) - Please choose one

MVC stands for -----

▶ **Model View Controller**

- ▶ Modern View Center
- ▶ Model View Center
- ▶ Modern View Controller

Question No: 5 (Marks: 1) - Please choose one

A -----
is a code that explains itself without the need of comments and extraneous documentation

- ▶ **Self documenting code**
- ▶ Self telling Code
- ▶ Self Documenting Design
- ▶ Non of the above

Question No: 6 (Marks: 1) - Please choose one

Bugs that won't "stand still" (almost random) are the ----- to deal with.

- ▶ least difficult
- ▶ **most difficult**
- ▶ very easy
- ▶ easy

Question No: 7 (Marks: 1) Please choose one **In refining**
the DFD during transaction mapping it is unnecessary to create a PSPEC since only the CS
PEC is relevant to this type of architectural style.

- ▶ True
- ▶ **False**

Question No: 8 (Marks: 1) Please choose one

Association is a type _____ of relation

- ▶ **Weak**
- ▶ Stronger
- ▶ Normal
- ▶ None of above

Question No: 9 (Marks: 1) Please choose one

Variable names must be in mixed case starting with upper case.

- ▶ True
- ▶ **False**

Question No: 10 (Marks: 1) - Please choose one

Bugs that won't

▶ **stand still**

▶ Dynamic

▶ Static

▶ None of above

(almost random) are the ----- to deal with.

Question No: 11 (Marks: 1) Please choose one
udefollowing:-

CRUD incl

▶ Creat

▶ Update

▶ Read

▶ **All of above**

Question No: 12 (Marks: 1) - Please choose one

Bugs Finding is done by which of the teams in Software Development lifecycle?

▶ Development Team

▶ **Testing Team**

▶ Analysis & Design Team

▶ Process Team

Question No: 13 (Marks: 1) Please choose one

Software

Bugs havemultiple names, which one of the below is not the name of Software Bugs -----
-

▶ Bugs

▶ Defects

▶ Errors

▶ **Mistakes**

Question No: 14 (Marks: 1) - Please choose one

In the debugging process a stack trace is a very useful tool.

▶ **True**

▶ False

Question No: 15 (Marks: Please choose one

A ----- is a code that explains itself without the need of comments and extraneous documentation

► **Self documenting code**

- Self telling Code
- Self Documenting Design
- None of the given

Question No: 16 (Marks: 1) Please choose one Project -----
---defines the concept and range of the proposed solution, a

limitations identify certain capabilities that the product will not include

► **Scope**

- Agreement
- Plan
- None of the given

Question No: 17 (Marks: 1) Please choose one In the N-
TireArchitecture the idea is to enhance scalability and -----
- by distributing both the data and the application using multiple server machines.

► **Performance**

- Efficiency
- Usability
- non of the all

Question No: 18 (Marks: 1)

The cyclomatic complexity metric provides the designer with information regarding the number of

- Cycles in the program
- Errors in the program
- **Independent logic paths in the program**
- Statements in the program

Question No: 19 (Marks: 1)

Which of the followings is not a testing type?

- Subsystem testing
- Alpha testing
- Beta Testing

► Gamma Testing

Question No: 20 (Marks: 1)

the items listed below is not one of the software engineering layers?

Which of

► Process

► Manufacturing

► Methods

► Tools

Question No: 21 (Marks: 1)

Which piece of code is self-documented?

► if (x==0) // this is the case when we are allocating a new number

► if (AllocFlag == 0)

► If (AllocFlag == NEW_NUMBER)

► None of the given

Question No: 22 (Marks: 2)

Define Modularity.

Question No: 23 (Marks: 2)

ate between Architectural Design and System Architecture in one line.

Differenti

Question No: 24 (Marks: 2)

What is meant by Software Debugging?

Question No: 25 (Marks: 2)

ments were given which required True or False with respect to

Five state

Equivalence Class (Statements I don't Remember :P)

Question No: 26 (Marks: 3)

ments are required to be indented and how it should be done, explain with

Why com

one example?

Question No: 27 (Marks: 3)

What are the different phases of testing? Draw a diagram. (No write up required)

Question No: 28 (Marks: 5)

Why and how complex expressions should be written in multiple short statements, Explain it with example

Question No: 29 (Marks: 5)

What is the Usefulness of Testing?

Question No: 30 (Marks: 5)

What are the static analyzers, give a check list of the requirements?

FINAL TERM EXAMINATION

Spring 2010

CS504- Software Engineering - I (Session - 2)

Time: 90 min

Marks: 58

Question No: 1 (Marks: 1) - Please choose one

first step in any OOA process model is to _____

- ▶ build an object-relationship model
- ▶ define collaborations between objects
- ▶ elicit customer requirements
- ▶ select a representation language

Question No: 2 (Marks: 1) - Please choose one

The

A
necessary supplement to transform or transaction mapping needed to create a complete architectural design
is _____.

- ▶ entity relationship diagrams
- ▶ the data dictionary
- ▶ processing narratives for each module
- ▶ test cases for each module

Question No: 3 (Marks: 1) - Please choose one

What
types of errors are missed by black-box testing and can be uncovered by
white-box testing?

- ▶ behavioral errors
- ▶ subtle logic errors
- ▶ performance errors
- ▶ input error

Question No: 4 (Marks: 1) - Please choose one

Whi
ch test criteria should be applied in a phase of testing?

- ▶ functional validity
- ▶ interface integrity
- ▶ correctness
- ▶ all of the given

Question No: 5 (Marks: 1) - Please choose one

The
hardest single part of building a software system is deciding precisely ----- to
build.

- ▶ what
- ▶ How
- ▶ When

- ▶ Why

Question No: 6 (Marks: 1) - Please choose one

In sequence Diagrams objects are organized in a -----line

- ▶ horizontal
- ▶ vertical
- ▶ horizontal and vertical
- ▶ non of the above

Question No: 7 (Marks: 1) - Please choose one

Every view is potentially Hierarchical in this regard which of the view are Hierarchical ?

- ▶ Function View
- ▶ Development View
- ▶ Concurrency view
- ▶ All of given

Question No: 8 (Marks: 1) - Please choose one

Like analysis models, many different kinds of -----models are developed

- ▶ Architectural
- ▶ System
- ▶ Design
- ▶ Logical

Question No: 9 (Marks: 1) - Please choose one

behind zero install architecture is to develop a system where no installation on the ----- is needed. idea

- ▶ Client side
- ▶ Server Side
- ▶ Client & Server Side
- ▶ Non of the all

Question No: 10 (Marks: 1) - Please choose one

The construction is used to avoid ----- errors.

- ▶ compilation
- ▶ runtime
- ▶ design time
- ▶ non of them

Question No: 11 (Marks: 1) - Please choose one

All

comments should be written in-----

- ▶ English
- ▶ French
- ▶ C++
- ▶ JAVA

Question No: 12 (Marks: 1) - Please choose one

One of the Guidelines to avoid common mistakes is to never use ----- except for declaration

- ▶ ,
- ▶ ;
- ▶ =
- ▶ ||

Question No: 13 (Marks: 1) - Please choose one

The

order in which bytes of one word are stored is ----- dependent

- ▶ Hardware
- ▶ Software
- ▶ language
- ▶ system

Question No: 14 (Marks: 1) - Please choose one

Verif

ication and validation are the processes in which we check a product against its

----- and the ----- of the users who will be using it

- ▶ specifications, expectations
- ▶ statements, expectations
- ▶ specifications , arguments
- ▶ statements, arguments

Question No: 15 (Marks: 1) - Please choose one

Two

tests are considered to be equivalent if it is believed that: if one discovers a defect, the other probably will too, and if one does not discover a defect, -----

- ▶ the other probably may
- ▶ non of them
- ▶ the other probably won't either
- ▶ the other probably will

Question No: 16 (Marks: 1) - Please choose one

It was lady named ----- who actually coin the term "bug" for the fiest time.

- ▶ Elisay Chistopher
- ▶ Admiral Grace Hopper
- ▶ Ana Nicholson
- ▶ Jane Hopper

Question No: 17 (Marks: 1) - Please choose one

The

first "bug" was actually a moth, which flew through an open window and into one of the Mark ----- 's relays.

- ▶ II
- ▶ I
- ▶ III
- ▶ IV

Question No: 18 (Marks: 1) - Please choose one

Sym

ptoms of logical errors are -----

- ▶ code is misbehaving
- ▶ The program doesn't crash, but akes odd branches through the code.
- ▶ Results are the opposite
- ▶ Output looks strange,
- ▶ all of the given

Question No: 19 (Marks: 1) - Please choose one

First

hand accounts of the problem are always useful in Debugging process.

- ▶ True
- ▶ False

Question No: 20 (Marks: 1) - Please choose one

Soft

ware architecture elements are further divided into categories which in total are -----

- ▶ 2
- ▶ 3
- ▶ 4
- ▶ 5

Question No: 21 (Marks: 1) - Please choose one

Esta

blishing responsibilities for objects includes

- ▶ Generalization Relationships
- ▶ Specialization Relationships
- ▶ all of the above
- ▶ identifying Association relationships

Question No: 22 (Marks: 1) - Please choose one

Sour

ce files can have the extension -----

- ▶ c++ .
- ▶ .C
- ▶ .cpp
- ▶ all of the given

Question No: 23 (Marks: 1) - Please choose one

Be very careful when you use functions with side effects – functions that change the values of the

- ▶ parameters
- ▶ variables
- ▶ constants
- ▶ None of given

Question No: 24 (Marks: 1) - Please choose one

Whi

ch of the following is not a characteristic of software?

- ▶ Software is tangible
- ▶ A change to a piece of code may implicitly affect the functions of the rest of the code.
- ▶ Software is configurable.
- ▶ Software does not wear and tear.

Question No: 25 (Marks: 1) - Please choose one

Whi

ch of the following is a fact finding method?

- ▶ Site visits
- ▶ Prototyping
- ▶ Study of similar systems
- ▶ All of given

Question No: 26 (Marks: 1) - Please choose one

The

statements given below are associated with system development, testing and maintenance. Identify the correct statement from among them:

- ▶ Some of the activities in the implementation phase are building individual system components, writing of programs and development of user interfaces.
- ▶ The post implementation review is performed to see that the computer system is working.

► None of Given

► System development phase consists of a development phase and an implementation phase.

Question No: 27 (Marks: 2)

What is Software Testing?

Question No: 28 (Marks: 2)

Write unit testing qualitative benefits.

Question No: 29 (Marks: 2)

Where Term Compute can be used in Methods?

Question No: 30 (Marks: 2)

Which of the following is a/are non-functional requirement of an e-learning system?

- (a) User friendliness
- (b) Time taken to download study materials through the system should not be too long.
- (c) On-line assignment submission facility
- (d) On-line chatting facility
- (e) Robustness

Question No: 31 (Marks: 3)

Write unit testing principles.

Question No: 32 (Marks: 3)

use of do while loops should be avoided. Explain why ?

The

Question No: 33 (Marks: 3)

3 Equivalence partitioning guidelines

Give

Question No: 34 (Marks: 5)

us the symptoms and an example of memory overrun bug class.

Disc

Question No: 35 (Marks: 5)

e at least 5 General Naming conventions for C++ or Java

Writ

Question No: 36 (Marks: 5)

lin at least 2 code structures.

Expa

FINAL TERM EXAMINATION

Spring 2010

CS504- Software Engineering - I (Session - 3)

Time: 90 min

Marks: 58

Question No: 1 (Marks: 1) - Please choose one

Quantitative methods for assessing the quality of proposed architectural designs

are readily available.

- ▶ True
- ▶ **False**

Question No: 2 (Marks: 1) - Please choose one

decision table should be used_____

A

- ▶ to document all conditional statements
- ▶ to guide the development of the project management plan
- ▶ only when building an expert system
- ▶ **when a complex set of conditions and actions appears in a component**

Question No: 3 (Marks: 1) - Please choose one

cases should be designed long before testing begins.

Test

- ▶ True
- ▶ **False**

Question No: 4 (Marks: 1) - Please choose one

Which of the following are characteristics of testable software?

- ▶ observability
- ▶ simplicity
- ▶ stability
- ▶ **all of the given**

Question No: 5 (Marks: 1) - Please choose one

Comparison testing is typically done to test two competing products as part of customer market analysis prior to product release.

- ▶ True
- ▶ **False**

Question No: 6 (Marks: 1) - Please choose one

collecting software metrics and making use of existing software reliability models it is possible to develop meaningful guidelines for determining when software testing is done.

By

- ▶ **True**
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Configuration reviews are not needed if regression testing has been rigorously applied during software integration.

- ▶ True
- ▶ **False**

Question No: 8 (Marks: 1) - Please choose one

change becomes ----- because of close presence of data and functions

A

- ▶ **Localized**
- ▶ Private
- ▶ Global
- ▶ Accessible

Question No: 9 (Marks: 1) - Please choose one

Requirement engineering mainly deals with the ----- of the system

- ▶ **definition phase**
- ▶ development phase
- ▶ maintenance
- ▶ non of the above

Question No: 10 (Marks: 1) - Please choose one

60% of all defects found in software projects can be traced back to poor -----

40-

- ▶ **Requirements**
- ▶ Design
- ▶ Coding
- ▶ Testing

Question No: 11 (Marks: 1) - Please choose one

In the N-Tire Architecture the idea is to enhance scalability and ----- by distributing both the data and the application using multiple server machines.

- ▶ **Performance**
- ▶ Efficiency
- ▶ Usability
- ▶ non of the all

Question No: 12 (Marks: 1) - Please choose one

MVC pattern was based on the ----- pattern.

- ▶ **Observer**
- ▶ Structural
- ▶ Behavioral
- ▶ Non of them

Question No: 13 (Marks: 1) - Please choose one

Classes should be declared in individual header files with the file name matching the ----- name.

- ▶ **class**
- ▶ method
- ▶ object
- ▶ non of the all

Question No: 14 (Marks: 1) - Please choose one

Unit testing is roughly equivalent to ----- testing for hardware in which each chip is tested

thoroughly after manufacturing

- ▶ Circuit level
- ▶ **Chip Level**
- ▶ component level
- ▶ system level

Question No: 15 (Marks: 1) - Please choose one

Software should be tested more like hardware, with Built-in self testing: such that each unit can be tested

- ▶ freely
- ▶ **independently**
- ▶ dependtly
- ▶ completely

Question No: 16 (Marks: 1) - Please choose one

system ----- is the period in which tremendous pressure is on developers end to fix the problem and make the system running again A

- ▶ Uptime
- ▶ **Downtime**
- ▶ Currentime
- ▶ futuretimr

Question No: 17 (Marks: 1) - Please choose one

Which one of the given below is not a symptom of memory overrun?

- ▶ Program crashes quite regularly after a given routine is called, that routine should be examined for a possible overrun condition.

▶ If the routine in question does not appear to have any such problem the most likely cause is that another routine, called in the prior sequence, has already

trashed variables or memory blocks.

▶ Checking the trace log of the called routines leading up to one with the problem will often show up the error.

▶ **Compiler warnings.**

Question No: 18 (Marks: 1) - Please choose one

Software architecture elements are further divided into categories which in total are -----

▶ 2

▶ 3

▶ 4

▶ 5

Question No: 19 (Marks: 1) - Please choose one

Three tier architecture contains ----- layers

▶ Presentation

▶ Application

▶ Database

▶ **All of the above**

Question No: 20 (Marks: 1) - Please choose one

Variables should be initialized where they are declared and they should be declared in the ----- scope possible.

- ▶ **Smallest**
- ▶ largest
- ▶ medium
- ▶ None of the given

Question No: 21 (Marks: 1) - Please choose one

----- are two important tools that can help in managing and mastering the complexity of a program.

- ▶ **abstraction and encapsulation**
- ▶ abstraction and Inheritance
- ▶ abstraction and Polymorphism
- ▶ None of given

Question No: 22 (Marks: 1) - Please choose one

Which of the following is a/are tool used in requirement analysis?

- ▶ Flow Graphs
- ▶ **Data Flow Diagrams**
- ▶ Activity Networks
- ▶ Module Dependency Diagrams

Question No: 23 (Marks: 1) - Please choose one

Consider the following comment. “The software which I bought won’t run on windows and when it runs I can’t use WORD at the same time”. Which of the following do you think are violated by the newly bought software?

- ▶ dependability, interchangeability

- ▶ **platform independence, interoperability**
- ▶ reliability, dependency
- ▶ interoperability, reliability

Question No: 24 (Marks: 1) - Please choose one

..... is a diagramming technique used to identify the types of objects in the system and the static relationships that exist among them.

- ▶ Class Diagram
- ▶ Document flow diagrams
- ▶ Data flow diagrams
- ▶ **Flow charts**

Question No: 25 (Marks: 1) - Please choose one

Which of the following interaction style best suit the design of an interface for visually impaired users?

- ▶ **direct manipulation**
- ▶ direct manipulation
- ▶ natural language
- ▶ command line

Question No: 26 (Marks: 1) - Please choose one

Which of the following testing involve purely black box testing?

- ▶ unit testing, beta testing
- ▶ acceptance testing, interface testing
- ▶ beta testing, acceptance testing
- ▶ **integration testing, interface testing**

Question No: 27 (Marks: 2)

what does this mean” Object Creation and Life Time”?

Question No: 28 (Marks: 2)

one can avoid hazards caused by side effects while writing code. List the two guidelines.

How

Question No: 29 (Marks: 2)

What is the greatest advantage of exception handling?

Question No: 30 (Marks: 2)

2 Unit Testing Tips.

Give

Question No: 31 (Marks: 3)

Write unit testing quantitative benefits.

Question No: 32 (Marks: 3)

Comments should be indented relative to their position in the code? Give an example

How

Question No: 33 (Marks: 3)

Consider the following code fragment.

```
while a
{
while b
c
d
}
```

If you were to test this code, what would be the test technique to adopt?

Question No: 34 (Marks: 5)

Narrate the manner for the organization of Class and Interface declarations

Question No: 35 (Marks: 5)

Discuss the symptoms and an example of coding error bug class.

Question No: 36 (Marks: 5)

Code portability is so important ? Give out 3 ways / Guide lines to improve the code portability with examples

(5+5)

Why

FINALTERM EXAMINATION

Spring 2010

CS504- Software Engineering - I (Session - 4)

Time: 90 min

Marks: 58

Question No: 1 (Marks: 1) - Please choose one

are the three generic phases of software engineering?

What

- ▶ **definition, development, support**
- ▶ what, how, where
- ▶ programming, debugging, maintenance
- ▶ analysis, design, testing

Question No: 2 (Marks: 1) - Please choose one

the context of requirements analysis, partitioning results in the elaboration of data, function, or behavior.

In

► True

► False

Question No: 3 (Marks: 1) - Please choose one

Qua

ntitative methods for assessing the quality of proposed architectural designs are readily available.

► True

► False

Question No: 4 (Marks: 1) - Please choose one

In

refining the DFD during transaction mapping it is unnecessary to create a PSPEC since only the CSPEC is relevant to this type of architectural style.

► True

► False

Question No: 5 (Marks: 1) - Please choose one

In

transaction mapping the first level factoring results in the_____

► creation of a CFD

► derivation of the control hierarchy

► distribution of worker modules

► refinement of the module view

Question No: 6 (Marks: 1) - Please choose one

Vari

able names must be in mixed case starting with upper case.

► True

▶ False

Question No: 7 (Marks: 1) - Please choose one

Class

variables should be declared public. This concept violates which of the following.

▶ Information hiding

- ▶ Encapsulation
- ▶ Information hiding and Encapsulation
- ▶ None of given

Question No: 8 (Marks: 1) - Please choose one

Real

-time applications add a new and potentially difficult element to the testing mix

- ▶ performance
- ▶ reliability
- ▶ security

▶ time

Question No: 9 (Marks: 1) - Please choose one

Top-

down integration testing has as it's major advantage(s) that

- ▶ low level modules never need testing
- ▶ major decision points are tested early

► no stubs need to be written

► **none of the given**

Question No: 10 (Marks: 1) - Please choose one

Conf

Integration reviews are not needed if regression testing has been rigorously applied during software integration.

-

► **True**

► False

Question No: 11 (Marks: 1) - Please choose one

isAu

Authorization, assess Performance, get item for UPC, get cashier for number, are the examples of

► Services

► Attributes

► Links

► **Non of the above**

—

Question No: 12 (Marks: 1) - Please choose one

the d

Development view describes the -----organization of the software in its development environment,

► **Static**

- ▶ Dynamic
- ▶ Still
- ▶ Static & Dynamic

Question No: 13 (Marks: 1) - Please choose one

----- being the first pure Object Oriented language in which observer pattern was used in implementing its Model View Controller

- ▶ Smalltalk
- ▶ PASCAL
- ▶ **JAVA**
- ▶ C++

Question No: 14 (Marks: 1) - Please choose one

garian Notation was first discussed by Charles Simonyi of-----

▶ **Microsoft.**

- ▶ Oracle
- ▶ Apple Macintosh
- ▶ Non of the all

Question No: 15 (Marks: 1) - Please choose one

language does not specify whether ----- is signed or unsigned.

▶ **char**

- ▶ integer
- ▶ double
- ▶ constant

Question No: 16 (Marks: 1) - Please choose one

A

number of invisible execution paths can exist in simple code in a language that allows

▶ exceptions

- ▶ defects
- ▶ errors
- ▶ all of them

Question No: 17 (Marks: 1) - Please choose one

Bugs

Fixing is done by which of the teams in Software Development lifecycle?

▶ Development Team

- ▶ Testing Team
- ▶ Analysis & Design Team
- ▶ Process Team

Question No: 18 (Marks: 1) - Please choose one

Soft

ware Bugs have multiple names, Which one of the below is not the name of Software Bugs -----

▶ Bugs

- ▶ Defects
- ▶ Errors
- ▶ Mistakes

Question No: 19 (Marks: 1) - Please choose one

Sym

ptoms of logical errors are -----

- ▶ code is misbehaving

► The program doesn't crash, but takes odd branches through the code.

► Results are the opposite

► Output looks strange,

► **all of the given**

Question No: 20 (Marks: 1) - Please choose one

First

hand accounts of the problem are always useful in Debugging process.

► **True**

► False

Question No: 21 (Marks: 1) - Please choose one

In

the debugging process a stack trace is a very useful tool.

► **True**

► False

Question No: 22 (Marks: 1) - Please choose one

Esta

blishing responsibilities for objects includes

► Generalization Relationships

- ▶ Specialization Relationships
- ▶ all of the above

▶ identifying Association relationships

Question No: 23 (Marks: 1) - Please choose one

Stak

eholders are different people who would be interested in the -----

- ▶ Software
- ▶ System
- ▶ Product

▶ All of the given

Question No: 24 (Marks: 1) - Please choose one

A ---

----- is a code that explains itself without the need of comments and extraneous documentation

▶ Self documenting code

- ▶ Self telling Code
- ▶ Self Documenting Design
- ▶ None of the given

Question No: 25 (Marks: 1) - Please choose one

1.

Variables should be initialized where they are declared and they should be declared in the ----- scope possible.

▶ Smallest

- ▶ largest
- ▶ medium

- ▶ None of the given

Question No: 26 (Marks: 1) - Please choose one

Which of the following is an/are advantage(s) of object oriented analysis & design over structured system analysis & design?

- ▶ Ease of modeling real world events
- ▶ Reusability
- ▶ Maintainability

☒ **All of given**

Question No: 27 (Marks: 2)

What is called **self documenting code**.

Question No: 28 (Marks: 2)

What is an Inspection Checklist.

Question No: 29 (Marks: 2)

Give

2 examples of exceptional code pathes.

Question No: 30 (Marks: 2)

The

following written statement depicts which requirement from the requirement engineering process

“Constraints on the services or functions offered by the system such as timing constraints, constraints on the development process, standards, etc.”

Question No: 31 (Marks: 3)

Wri

te unit testing quantitative benefits.

Question No: 32 (Marks: 3)

Give

three general rules for avoiding split lines.

Question No: 33 (Marks: 3)

Expa

lin about 3 coverage schemes in white box testing.

Question No: 34 (Marks: 5)

List

five guidelines that can help you in writing portable code.

Question No: 35 (Marks: 5)

Belo

w is the chunk of code :

```
result = squareRoot(argument);  
assert (abs (result * result – argument) < epsilon);
```

Write the Contract for square root routine keeping in view unit testing.

Question No: 36 (Marks: 5)

Parentheses should always be used as they reduce complexity. Explain it with the help of a single example.

FINAL TERM EXAMINATION

Fall 2008

CS504- Software Engineering _ I (Session - 1)

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

Object-oriented domain analysis is concerned with the identification and specification of reusable capabilities within an application domain.

- True
- False

Question No: 2 (Marks: 1) - Please choose one

Fourth generation techniques

- Allow software to be developed without any testing.
- Eliminate the need for costly requirements gathering activities.
- Can reduce the time required to develop software.
- Are best used by non-programmers to build small systems.

Question No: 3 (Marks: 1) - Please choose one

Data design actually begins during the creation of the analysis model, not the architectural model.

- True
- False

Question No: 4 (Marks: 1) - Please choose one

The states shown in a state transition diagram do not necessarily correspond to the processes shown in a control flow diagram for the same system.

- True
- False

Question No: 5 (Marks: 1) - Please choose one

The criteria used to assess the quality of an architectural design should be based on system

- accessibility and reliability
- data and control
- functionality

- implementation details

Question No: 6 (Marks: 1) - Please choose one

A useful technique for evaluating the overall complexity of a proposed architecture is to look at the component

- number and size of components
- flow dependencies and sharing dependencies
- size and cost
- none of the given

Question No: 7 (Marks: 1) - Please choose one

The three basic principles that guide maintainability are: simplicity, clarity, and_____ .

- Generality
- Reliability

- All of the given choices
- None of the Given choices

Question No: 8 (Marks: 1) - Please choose one

In order to make a program self documented a number of attributes required. Which one is the attribute/s of self documented program

- All of the given choices
- Size of each function
- Choice of variable
- Modularity

Question No: 9 (Marks: 1) - Please choose one

Floating point constants should always be written with decimal point and at least

- one decimal
- two decimal
- three decimal
- none of the given

Question No: 10 (Marks: 1) - Please choose one

The code becomes self explanatory with the help of proper use of parentheses. Select the right one.

- leapYear = year % 4 == 0 && year % 100 != 0 || year % 400 == 0 ;
- leapYear = ((year % 4 == 0) && (year % 100 != 0) || ((year % 400 == 0)));
- leapYear = ((year % 4 == 0) && (year % 100 != 0)) || (year % 400 == 0);
- leapYear = (year % 4 == 0) && (year % 100 != 0) || (year % 400 == 0);
- none of the given options

Question No: 11 (Marks: 1) - Please choose one

Graph-based testing methods can only be used for object-oriented systems

- True
- False

Question No: 12 (Marks: 1) - Please choose one

Configuration reviews are not needed if regression testing has been rigorously applied during software integration.

- True
- False

Question No: 13 (Marks: 1) - Please choose one

A class is a ----- of objects

- Template
- Stereotype
- Collection
- Non of the above

Question No: 14 (Marks: 1) - Please choose one

A public Interface provides a way for with other Classes.

- Communication
- Accessibility
- Reaching
- All of the above

Question No: 15 (Marks: 1) - Please choose one

Flow Charts represents

- sequence Activity
- random activity
- parallel activity
- non of the above

Question No: 16 (Marks: 1) - Please choose one

Software architecture defines the high level structure of the software by putting together a number of architectural ----- in an organized fashion.

- elements
- parts
- components
- non of the all

Question No: 17 (Marks: 1) - Please choose one

Patterns are devices that allow programs to share knowledge about their -----.

- Design
- Code
- Analysis
- Non of the all

Question No: 18 (Marks: 1) - Please choose one

MVC stands for -----

- Model View Controller
- Modern View Center
- Model View Center
- Modern View Controller

Question No: 19 (Marks: 1) - Please choose one

A ----- is a code that explains itself without the need of comments and extraneous documentation

- Self documenting code
- Self telling Code
- Self Documenting Design
- Non of the them

Question No: 20 (Marks: 1) - Please choose one

Type conversions must always be done -----

- explicitly
- implicitly
- simultaneously
- non of them

Question No: 21 (Marks: 1) - Please choose one

Types that are -----to one file only can be declared inside that file.

- Local
- Global
- Private
- General

Question No: 22 (Marks: 1) - Please choose one

The use of *d*, *while* loops should be -----

- avoided
- encouraged
- practiced
- non of them

Question No: 23 (Marks: 1) - Please choose one

one of the causes of the portability issues is the order of ----- varies from one implementation to other.

- evaluation
- numbers
- variables
- symbols

Question No: 24 (Marks: 1) - Please choose one

----- is a powerful technique that separates error-handling code from normal code.

- Exception handling
- Code handling
- variable handling

- pointer handling

Question No: 25 (Marks: 1) - Please choose one

The goal of testing is to expose ----- defects in a software system before it is put to use.

- latent
- already present
- current
- runtime

Question No: 26 (Marks: 1) - Please choose one

Static analyzers are software tools for ----- processing.

- analysis text
- source text
- design text
- non of the given

Question No: 27 (Marks: 1) - Please choose one

The first bug was actually a moth, which flew through an open window and into one of the Mark ----- 's relays.

- II
- I
- III
- IV

Question No: 28 (Marks: 1) - Please choose one

Which one of the given below is not a symptom of memory overrun?

- Program crashes quite regularly after a given routine is called, that routine should be examined for a possible overrun condition.
- If the routine in question does not appear to have any such problem the most likely cause is that another routine, called in the prior sequence, has already trashed variables or memory blocks.
- Checking the trace log of the called routines leading up to one with the problem will often show up the error.
- Compiler warnings.

Question No: 29 (Marks: 1) - Please choose one

Holistic medicine, concerns itself with the state of the body as a whole, not the ----- that is currently attacking it.

- reason
- disease
- source
- non of the given

Question No: 30 (Marks: 1) - Please choose one

Bugs that won't "stand still" (almost random) are the ----- to deal with.

- least difficult

- most difficult
- very easy
- easy

ALL Papers Longs question some of them solved.....

FINAL TERM EXAMINATION

Spring 2009

CS504- Software Engineering _ I

Question No: 31 (Marks: 1)

Write the procedure how names representing Abbreviations and acronyms should be describe in coding style guide.

Abbreviations and acronyms should not be uppercase when used as name.

`exportHtmlSource(); // NOT: xportHTMLSource();`

`openDvdPlayer(); // NOT: openDVDPlayer();`

Question No: 32 (Marks: 1)

Define a system downtime.

A system downtime is the period in which tremendous pressure is on developers end to fix the problem and make the system running again.

Question No: 33 (Marks: 2)

What is layered architecture

As the name suggests, a layered architecture has many different layers. One typical example of a layered architecture is an operating system where different layers are used to provide services and functionality and the inner layers are closer to the machine hardware than the outer layers. In this way, each layer isolates the outer layer from inner complexities. In order to work properly, the outer layer only needs to know the interface provided by the inner layer. If there are any changes in the inner layer, as long as the interface does not change, the outer layer is not affected. This scheme tremendously portability of the system. The basic layered architecture is depicted in the following diagram.

Question No: 34 (Marks: 2)

What is Infeasible path?

Infeasible path is a path through a program which is never traversed for any input data.

Question No: 35 (Marks: 3)

List three guidelines that can help you in writing portable code.

Question No: 36 (Marks: 3)

Describe three coverage schemes related to white box testing.

Question No: 37 (Marks: 3)

Write the General Form for documenting patterns.

The motivation or context that this pattern applies to.

Prerequisites that should be satisfied before deciding to use a pattern.

A description of the program structure that the pattern will define.

A list of the participants needed to complete a pattern.

Consequences of using the pattern...both positive and negative.

Examples!

Question No: 38 (Marks: 5)

Discuss any five the General naming conventions for an object oriented language

General naming conventions for Java and C++

1. Names representing types must be nouns and written in mixed case starting with upper case.

Line, FilePrefix

2. Variable names must be in mixed case starting with lower case.

line, filePrefix

This makes variables easy to distinguish from types, and effectively resolves potential naming collision as in the declaration `Line line`;

3. Names representing constants must be all uppercase using underscore to separate words.

`MAX_ITERATIONS, COLOR_RED`

In general, the use of such constants should be minimized. In many cases implementing the value as a method is a better choice. This form is both easier to read, and it ensures a uniform interface towards class values.

`int getMaxIterations()` // NOT: `MAX_ITERATIONS = 25`

{

```
return 25;
```

```
}
```

4. Names representing methods and functions should be verbs and written in mixed case starting with lower case.

```
getName(), computeTotalWidth()
```

5. Names representing template types in C++ should be a single uppercase letter.

```
template<class T> ...
```

```
template<class C, class D> ...
```

6. Global variables in C++ should always be referred to by using the :: operator.

```
::mainWindow.open() , ::applicationContext.getName()
```

7. Private class variables should have _ suffix.

```
class SomeClass
```

```
{
```

```
private int length_;
```

```
...
```

```
}
```

Apart from its name and its type, the scope of a variable is its most important feature.

Indicating class scope by using _ makes it easy to distinguish class variables from local scratch variables.

8. Abbreviations and acronyms should not be uppercase when used as name.

```
exportHtmlSource(); // NOT: xportHTMLSource();
```

```
openDvdPlayer(); // NOT: openDVDPlayer();
```

Using all uppercase for the base name will give conflicts with the naming conventions given above. A variable of this type would have to be named dVD, hTML etc. which obviously is not very readable.

9. Generic variables should have the same name as their type.

```
void setTopic (Topic topic) // NOT: void setTopic (Topic value)
```

```
// NOT: void setTopic (Topic aTopic)
```

```
// NOT: void setTopic (Topic x)
```

```
void connect (Database database) // NOT: void connect (Database db)
```

// NOT: void connect (Database oracleDB)

Non-generic variables have a role. These variables can often be named by combining role and type:

Point startingPoint, centerPoint;

Name loginName;

10. All names should be written in English.

fileName; // NOT: filNavn

11. Variables with a large scope should have long names, variables with a small scope can have short names. Scratch variables used for temporary storage or indices are best kept short. A programmer reading such variables should be able to assume that its value is not used outside a few lines of code. Common scratch variables for integers are i, j, k, m, n and for characters c and d.

12. The name of the object is implicit, and should be avoided in a method name.

line.getLength(); // NOT: line.getLineLength();

The latter seems natural in the class declaration, but proves superfluous in use.

Question No: 39 (Marks: 5)

What are the Software testing objective? Also define a successful test.

Software testing objective

„h The correct approach to testing a scientific theory is not to try to verify it, but to seek to refute the theory. That is to prove that it has errors. (Popper 1965)

„h The goal of testing is to expose latent defects in a software system before it is put to use.

„h A software tester tries to break the system. The objective is to show the presence of a defect not the absence of it.

„h Testing cannot show the absence of a defect. It only increases your confidence in the software.

„h This is because exhaustive testing of software is not possible – it is simply too expansive and needs virtually infinite resources.

Successful Test

From the following sayings, a successful test can be defined

“If you think your task is to find problems then you will look harder for them than if you think your task is to verify that the program has none” – Myers 1979.

“A test is said to be successful if it discovers an error” – doctor’s analogy.

The success of a test depends upon the ability to discover a bug not in the ability to prove that the software does not have one. As, it is impossible to check all the different scenarios of a software application, however, we can apply techniques that can discover potential bugs from the application. Thus a test that helps in discovering a bug is a successful test. In software testing phase, our emphasis is on discovering all the major bugs that can be identified by running certain test scenarios. However it is important to keep in mind that testing activity has certain limitations.

Question No: 40 (Marks: 10)

What should be the sets of inputs that should be used to test the system effectively and efficiently? Give an example.

To answer these questions, we divide a problem domain in different classes. These are called Equivalence Classes.

Question No: 41 (Marks: 10)

Discuss Art and Science of Debugging.

Debugging is taken as an art but in fact it is a scientific process. As people learn about different defect types and come across situations in which they have to debug the code, they develop certain heuristics. Next time they come across a similar situation, they apply those heuristics and solve the problem in lesser time and with a lesser effort. While discussing the debugging process we discuss the phenomenon of “you miss the obvious”.

When a person writes a code, he develops certain impression about that code. One can term this impression as a personal bias that the developer builds towards his creation the

“code” and when he has to check this code, he can potentially miss out obvious mistakes due to this impression or bias. Therefore, it is strongly recommended that in order to reach to a defect in the code, one needs “another pair of eyes”. That is, start discovering the defect by applying your own heuristics and if you could reach to the problem, fine, otherwise ask a companion to help you in this process. We shall further elaborate this

idea based on the following example.

Final Term Examination - February 2005

Time Allowed: 150 Minutes

Question No. 1

Marks : 6

What are Loop Errors? Describe briefly. What are symptoms of Loop Errors?

Question No. 2 Marks : 15

Loop Errors

- „h Loop errors break down into several different subtypes.
- „h They occur around a loop construct in a program.
- „h Infinite loops, off-by-one loops, and improperly exited loops.

Symptoms

- „h If your program simply locks up, repeatedly displays the same data over and over, or infinitely displays the same message box, you should immediately suspect an infinite loop error.
 - „h Off-by-one loop errors are quite often seen in processes that perform calculations.
 - „h If a hand calculation shows that the total or final sum is incorrect by the last data point, you can quickly surmise that an off-by-one loop error is to blame.
 - „h Likewise, if you were using graphics software and saw all of the points on the screen, but the last two were unconnected, you would suspect an off-by-one error.
- Watching for a process that terminates unexpectedly when it should have continued.

Q- 4

Marks: 6

What are Boolean Bugs? Describe briefly. What are symptoms of Boolean Bugs?

Boolean bugs

Boolean bugs occur because the mathematical precision of Boolean algebra has virtually nothing to do with equivalent English words.

When we say "and", we really mean the boolean "or" and vice versa.

Symptoms

„h When the program does exactly the opposite of what you expect it to. For example, you might have thought you needed to select only one entry from a list in order to proceed. Instead, the program will not continue until you select more than one. Worse, it keeps telling you to select only one value.

„h For true/false problems, you will usually see some sort of debug output indicating an error in a function, only to see the calling function proceed as though the problem had not occurred.

2. Write a short note on Data Flow or Pipes and Filter Architecture Configuration.(3 Pts)

This architecture is very similar to data flow diagrams. This is used when the input data is processed through a series of transformations to yield the desired output. It is also known as pipes and filters architecture where each processing step is called a filter and the connecting link from one process to the other is called the pipe through which the information flows from one process to the other. An important aspect of this model is that each filter works independently of others and does not require knowledge of the working of any of the other filters including its neighbours.

If the dataflow has only a single sequence of processes with no alternative or parallel paths, then it is called batch sequential. These models are depicted in the following diagrams.

3. How do you differentiate between the responsibilities of Developer and Tester? (3 Pts)

Development is a creative activity Testing is a destructive activity

Objective of development is to show that the program works. Objective of testing is to show that the program does not work

Final Term Examination - August 2004

QuestionNo.3**Marks:6**

What are the symptoms of pointer errors?

Pointer errors

„h A pointer error is any case where something is being used as an indirect pointer to another item.

„h Uninitialized pointers: These are pointers that are used to point at something, but we fail to ever assign them something to point at.

„h Deleted pointers, which continue to be used.

„h An Invalid pointer is something that is pointing to a valid block of memory, but that memory does not contain the data you expect it to.

Symptoms

„h The program usually crashes or behaves in an unpredictable and baffling way.

„h You will generally observe stack corruptions, failure to allocate memory, and odd changing of variable values.

„h Changing a single line of code can change where the problem occurs.

„h If the problem "goes away" when you place a print statement or new variable into the code that you suspect contains the problem.

What are the differences between Thin Client and Fat Client architecture?

(Q 9)

Thin Client Model

This model was initially used to migrate legacy systems to client server architectures.

In this case the legacy system may act as a server in its own right and the GUI may be implemented on a client. Its chief disadvantage is that it places a heavy processing load on both the server and the network.

Fat Client Model

With advent of cheaper and more powerful hardware, people thought of using the processing power of client side machines. So the fat client model came into being. In

this model, more processing is delegated to the client as the application processing is locally extended. It is suitable for new client/server systems when the client system capabilities are known in advance. It however is more complex than thin client model with respect to management issues. Since the client machine now also has a significant part of the application resident on it, new versions of each application need to be installed on every client.

Following is the list of bugs' symptoms; identify the bug classes in each case.

- a) System slowdowns. Memory Leakage
- b) The program doesn't crash, but the flow of the program takes odd branches through the code. Logical Error
- c) If your program simply locks up, repeatedly displays the same data over and over, or infinitely displays the same message box. Loop Error

(GOOD LUCK FOR FINAL EXAM)
(KEEP VISITING VURANK FOR MORE)